

ADITYA KULKARNI

(281) 876-7066 | adityakulkarnius@gmail.com | linkedin.com/in/adityakulkarni244 | github.com/Aditya-k24 | kulkarniaditya.com

SUMMARY

AI-systems engineer (MS CS, NC State, May 2026) who builds **LLM infrastructure** other engineers consume: **MCP** tool servers, multi-model routing, and durable agent orchestration. Strong in **Python**, **TypeScript**, and **Java**, with production **ML** pipelines and Kubernetes-native cloud deployment experience.

EXPERIENCE

NC State University

Graduate Research Assistant

Apr 2025 – Present

Raleigh, NC

- Designed a cloud-based genomic platform (**JBrowse 2** on **Google Cloud Run**) enabling **100+ researchers** worldwide to interactively explore **500GB** of strawberry and grape genome datasets
- Collaborated on **ML** architecture design, applying hyperparameter tuning and time-series analysis to improve model accuracy by **12%**

Isomer AI

Product Engineer Intern

May 2025 – May 2026

Remote

- Built **MCP servers** in **Python** and **TypeScript** exposing typed tools and context to multi-model **LLM** agents (Claude, GPT-4o, Gemini), delivering **AI as a service** to product teams
- Engineered a **Temporal.io** workflow engine for durable agent runs: versioned Draft/Live/Archived releases, canary traffic splits, QA sampling, and CRITICAL/HIGH/STANDARD risk detection
- Shipped **10+ features** end-to-end (**PostgreSQL** to **React.js**) with **Playwright** browser automation over Tailscale VPN, **Auth0** RBAC, **Redis** caching, and **Sentry** observability across per-tenant stacks

Mphasis (HP Company)

Software Developer Intern

Jun 2024 – Aug 2024

Bangalore, India

- Built **5+ AI workflows**, including a health-claim automation tool using **OCR**, **NLP**, and **ML** classification, that improved automated claim accuracy by **25%**
- Automated deployments on **AWS** and **GCP** with **Grafana** monitoring dashboards, and supported model validation and red-teaming to enhance production ML reliability

Capgemini

AI/ML Intern

Jun 2023 – Aug 2023

Mumbai, India

- Led the SDLC for a support-ticket automation system, cutting manual processing time **15%**; built **Hadoop**, **Spark**, and **Kafka** pipelines that reduced processing latency **20%**

PROJECTS

CortexQ | Kubernetes LLM Inference Platform

2026

- Built a **Kubernetes**-native inference platform: **FastAPI** router with a 3-state circuit breaker and **EWMA** latency-aware load balancing across Claude, GPT-4o, and Gemini; **KEDA** autoscaler scales pods 0→N on **Redis** queue depth (scale-to-zero when idle)
- Authored a **Python** Kopf CRD operator reconciling Deployments, Services, and ScaledObjects; **OpenTelemetry** + Prometheus + Grafana observability; **Helm** + **ArgoCD** GitOps; 88 pytest tests; 500 VU k6 load tests

LLM Eval Harness | Vendor-Neutral Multi-Model Benchmark

2025

- Benchmarked Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across grounded QA, multi-hop QA, and claim verification on a **2,000-example** frozen set with 3 repeated runs per model
- Tracked ExactMatch, TokenF1, grounded hallucination rate, p50/p95 latency, and cost/call; established significance via paired bootstrap CIs, McNemar test, and Holm correction

VectorLift | Semantic Search & Ranking Engine (RAG)

2026

- Built a semantic search system on **MS MARCO**: **BM25** baseline, fine-tuned dense bi-encoder, and cross-encoder re-ranker; **Kafka** indexing into **Elasticsearch** + **Qdrant**; served via **FastAPI**; tuned with **NDCG@10/MRR@10**

PulseRank | Real-Time Leaderboard & Activity Feed

2026

- Architected an event-driven leaderboard with **sub-millisecond Redis** reads, a **MySQL** outbox for durable writes, and live updates over Server-Sent Events; **TypeScript**, **NestJS**, **Next.js**
- Verified multi-store consistency with **Testcontainers** integration tests, instrumented with **OpenTelemetry**, and load-tested throughput with **k6** benchmarks

EDUCATION

NC State University

M.S. Computer Science

Aug 2024 – May 2026

Raleigh, NC

- Relevant Coursework:** Artificial Intelligence, Machine Learning, Algorithms, Operating Systems, Software Engineering, Data Science

Mumbai University

B.Tech Computer Science, CGPA 9.32/10

Aug 2020 – May 2024

Mumbai, India

- Relevant Coursework:** Data Structures & Algorithms, Big Data Infrastructure, Data Warehousing, Blockchain, Software Engineering

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, Ruby, SQL

AI / ML: LLM APIs (Claude, GPT-4o, Gemini), MCP, RAG, vector search (Qdrant), PyTorch, sentence-transformers

Frameworks: FastAPI, Node.js, Express.js, React.js, Next.js, NestJS, Spring Boot

Infra & Tools: Kubernetes, Docker, KEDA, ArgoCD, Helm, Temporal.io, Kafka, AWS, GCP, Playwright, Grafana

Achievements: HackNC24 Top 10; Sanjeevani Top 1% of 1100+ teams; IEEE-published; 110+ RSpec tests on Expertiza Rails OSS